# Software Design Description 000

## ## Total Component Summary

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# Development Standards

## Code Organization

1. Modular file system - multiple concise files (<300 lines ideally), not monolithic main files
2. One file, one purpose - minimal global interfaces per file. **NO GLOBAL VARIABLES**
3. Proper file separation with corresponding headers

## Architecture Principles

- File-scope coupling: Static functions may be tightly integrated for performance
- Module-scope coupling: Loosely coupled between files via minimal interfaces
- Strong cohesion within modules
- No global variables - use proper encapsulation

## Documentation

- Document pin allocations and feature implementations
- Update relevant README.md or commit messages
- Headers on files for clarity
- Commit frequently

## Standards Compliance

- Adhere to Unix philosophy where applicable
- Follow C standards for embedded code
- Follow Verilog standards for FPGA code
- Follow Dart/Flutter standards for app development
- Use ESP-IDF and FreeRTOS best practices

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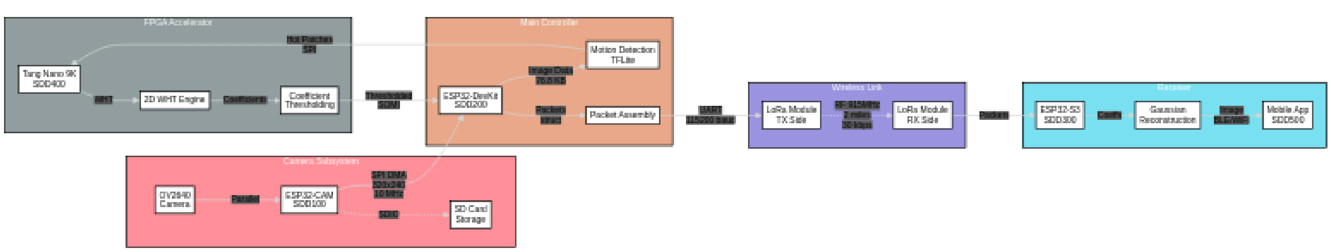


## ESP32-CAM:

- Capture, store, and pass images rapidly to ESP32-DEVKITV1

### ESP32-DEVKITV1:

- Serve as intermediary between CAM and FPGA.
- House (back end) edge ML model for motion prediction.
- Manage stepper motor, RTC, and temperature/humidity sensor.
- Manage pre-rendered backgrounds for TFT UI.

### Tang Nano 9K:

- Manage TFT display interfacing (480x272 RGB LCD).
- WHT compression pipeline: 2D Walsh-Hadamard Transform on 8x8 blocks, sequency masking, thresholding, quantization (SDD400/SDD401).
- PSRAM frame buffer (64 Mbit embedded) for frame storage and diff computation.
- I2C GPIO expander for stepper motor control and joystick buttons.
- Return compressed coefficients to DevKitV1 via SOMI (SPI MISO).

### ESP32-S3 (WROOM):

- Receive WHT coefficients via LoRa.
- Gaussian reconstruction: inverse WHT + smoothing to reconstruct frames from diffs.
- WiFi AP ("TrailCamera") + WebSocket server for mobile app streaming.
- Reference frame storage in PSRAM (3 positions x 307 KB).

### iPhone:

- Final virtual user interface.
- Polish and patch partitioned ML reconstructed image data.
- On/Off, motor control, training en/dis.

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# References

## Other Documents

- Camera Implementation
- WiFi Implementation
- Stepper Motor Implementation

## Vendor Tools

- esp-idf
- camera component

## Definitions

- Cohiesion and Coupling
- Composability
- Unix Design Philosophy – The authors of C
- FreeRTOS

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# Relevant Standards and Practices

- Tightly coupled functions at static (file) scope
    - Simple and composable
- Loosely coupled functions when globally scoped

- Strong cohesion overall
- Use of ESP supported software components where possible
- Proper use of file seperation
  - Minimal global interfaces per file
  - One file has one purpose
- Proper use of FreeRTOS